

Drum brake - design brief

AI BRAKING / AB-DR / C0 / 2026-09-07

A spring-applied, thruster-released 315 mm drum concept with replaceable shoes, accessible pivots and a fabricated base. DIN compatibility is a future interface check, not claimed by this geometry.

TARGETS (not validated)

Drum diameter: 315 mm. Modelled geometry

Friction width: 100 mm. Modelled geometry

Study holding torque: 500 N·m. Unverified requirements hypothesis

Assumed minimum friction: 0.30. Sensitivity input, not lining data

Required total shoe normal force: 10.58 kN. $T / (\mu r)$; excludes self-energization

DESIGN DECISIONS

Buy a qualified thruster first; put development effort into accessible metalwork and service interfaces.

Keep pivot pins and bushes separately replaceable rather than boring directly into welded levers.

Use a fabricated prototype base; casting economics follow actual batch volumes.

CANDIDATE MATERIALS

Base and lever plate: S355J2+N candidate; weld procedure and fatigue detail approval required.

Pins: 42CrMo4 candidate with controlled finish; bushes: purchased bearing-grade bronze candidate.

Lining: bought, characterized non-asbestos compound; bonding and mechanical retention require supplier approval.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Locate base on the test fixture and inspect pivot-centre coordinates.
- 2. Fit clevis supports, levers, bushes and retained pivot pins.
- 3. Install carriers and lining; centre the drum and establish cold clearance.
- 4. Install purchased thruster and engineered equalizing/spring linkage after kinematic approval.
- 5. Check full release motion without friction contact before load testing.

OPEN DESIGN RISKS

- 1. Equalization, spring linkage and actuator transmission are layout placeholders; motion is not solved.
- 2. The assumed torque is not a rating and does not cover an emergency stop or hoist safety case.
- 3. Shoe contact geometry, drum heating and pivot loads require coupled assessment.

VALIDATION REQUIRED

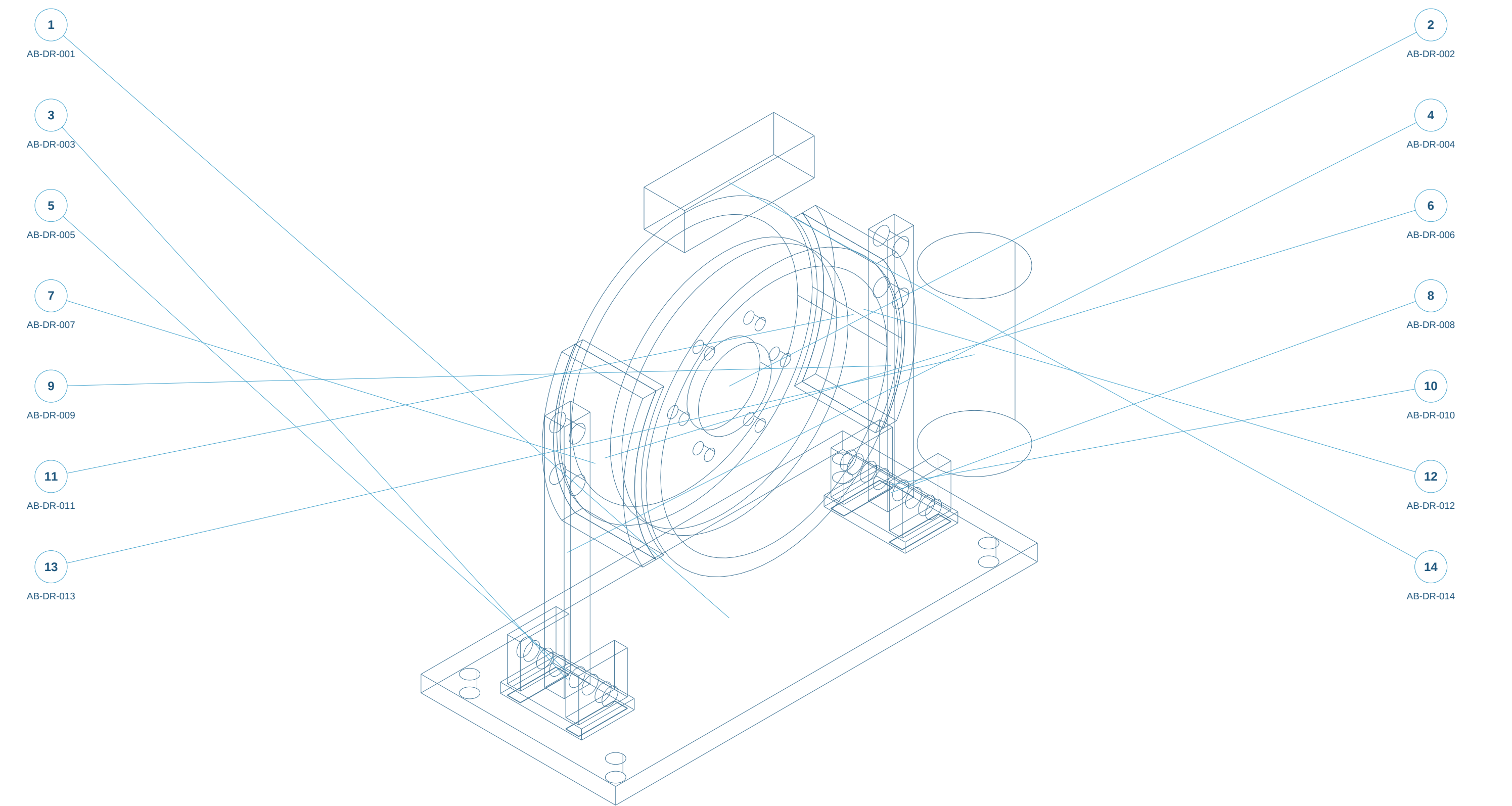
- 1. Kinematic travel/clearance and load transfer before structural FEA.
- 2. Lever/pin bearing, bending, base weld fatigue and drum thermal calculations.
- 3. Static holding, dynamic torque, application/release time and endurance tests with the chosen lining.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Advisory position / wear sensing with replaceable brackets; no AI-controlled brake release.
- 2. Record supplier sensor range, supply, connector, EMC, ingress and temperature limits before selection.
- 3. Define independent machine safety interface and diagnostic fault response with the responsible controls engineer.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-DR-001 | Four-hole base | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DR-002 | Study brake drum | make-concept

C45 prototype candidate; friction/thermal qualification required; Saw/laser blank; CNC critical faces and bores

AB-DR-003 | Pivot clevis -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DR-004 | Lever -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DR-005 | Lower pivot pin -1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; specify retention after review

AB-DR-006 | Lining -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy friction lining / machine carrier to approved retention drawing

AB-DR-007 | Shoe carrier -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Buy friction lining / machine carrier to approved retention drawing

AB-DR-008 | Pivot clevis 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DR-009 | Lever 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-DR-010 | Lower pivot pin 1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; specify retention after review

AB-DR-011 | Lining 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy friction lining / machine carrier to approved retention drawing

AB-DR-012 | Shoe carrier 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Buy friction lining / machine carrier to approved retention drawing

AB-DR-013 | Thruster reserved space | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy selected thruster; replace with vendor STEP

Parts continued

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AB-DR-014 | Spring/equalizer reserved space | interface-envelope

Supplier-controlled; illustrative reserved envelope; Engineer spring/linkage after kinematics

AB-DR-001 / Four-hole base

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 520.00 / 240.00 / 20.00 mm

520 x 240 x 20; 4 x diameter 18 THRU.

Hole centres X +/-230, Y +/-90 from part origin.

Machine base datum; mounting preload unassigned.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

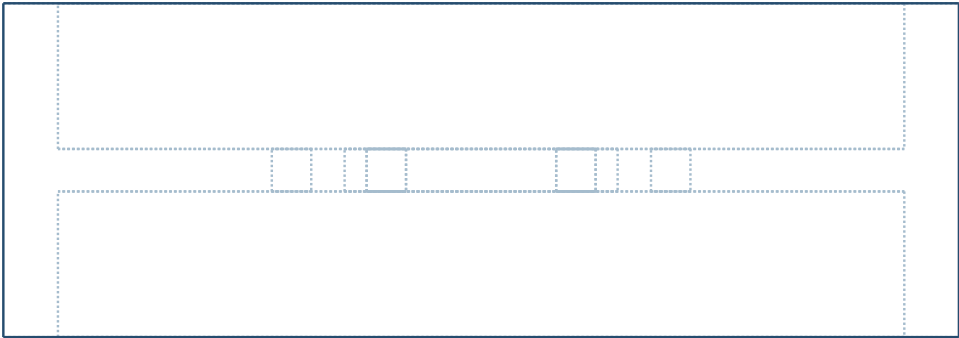
SIDE Y-Z



AB-DR-002 / Study brake drum

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TOP X-Y



C45 prototype candidate; friction/thermal qualification required

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 315.00 / 110.00 / 315.00 mm

OD 315; friction width 100 within 110 mm axial envelope.

Web bore 90; 6 x diameter 13 on PCD 125.

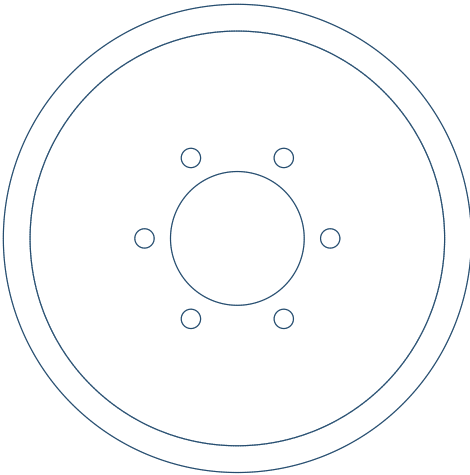
Shaft hub and retention not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

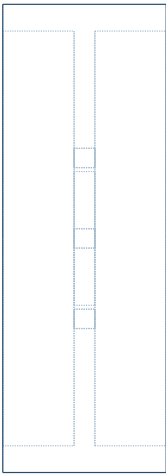
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



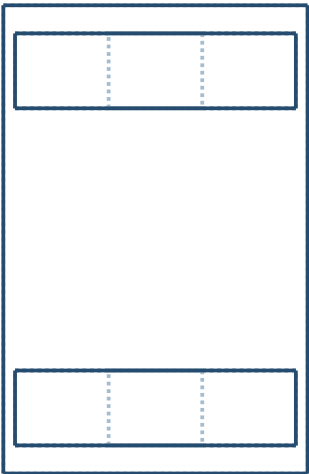
SIDE Y-Z



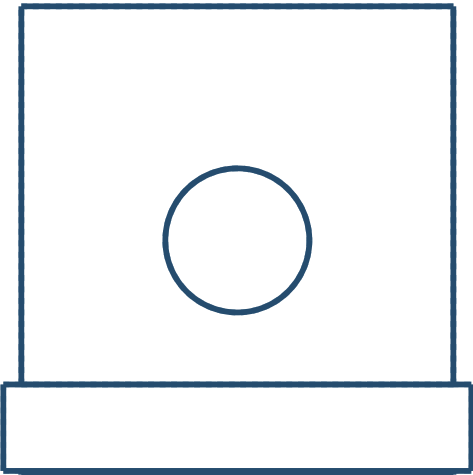
AB-DR-003 / Pivot clevis -1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 65.00 / 100.00 / 65.00 mm

Clevis pin bore 20; nominal ear gap 56 mm.

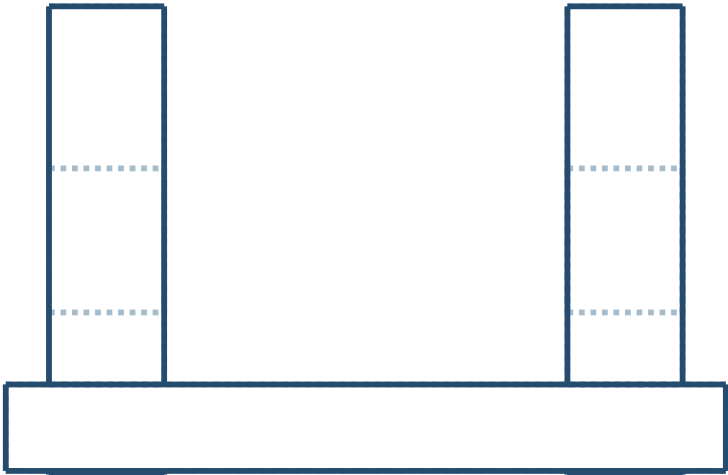
Foot 4 x diameter 9 at (+/-20,+/-35); base mating holes pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

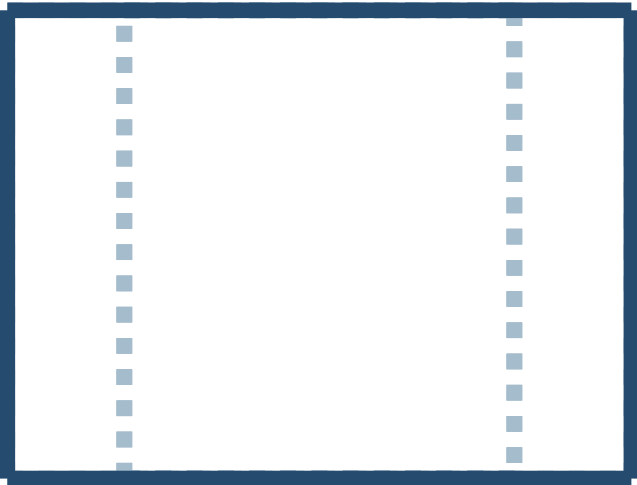
SIDE Y-Z



AB-DR-004 / Lever -1

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TOP X-Y



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 32.00 / 24.00 / 290.00 mm

32 x 24 section; pin centres Z 0, 205, 260 in local coordinates.

Nominal pin bore 20; bush fit not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



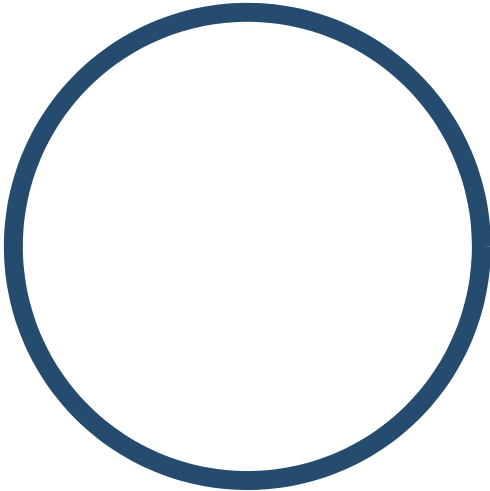
AB-DR-005 / Lower pivot pin -1

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TOP X-Y



FRONT X-Z



42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind; specify retention after review

Overall X/Y/Z bounds: 19.80 / 105.00 / 19.80 mm

Diameter 19.8 study clearance in nominal 20 bore.

No final fit, hardness or axial retainer design.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

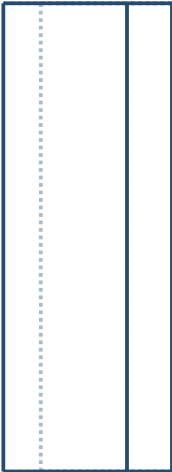
SIDE Y-Z



AB-DR-006 / Lining -1

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 36.03 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

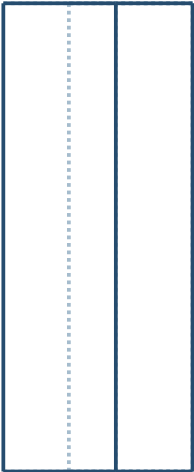
SIDE Y-Z



AB-DR-007 / Shoe carrier -1

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 40.42 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

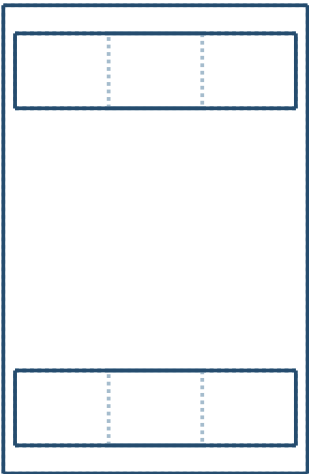
SIDE Y-Z



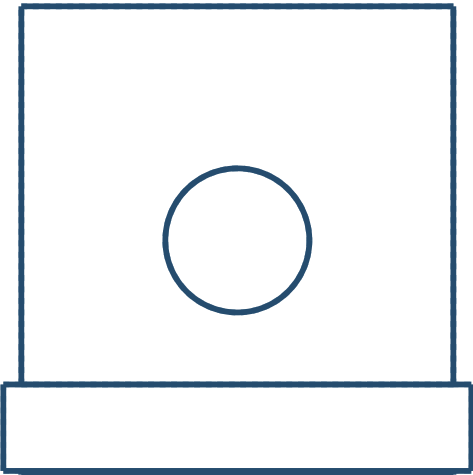
AB-DR-008 / Pivot clevis 1

AI BRAKING / AB-DR / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 65.00 / 100.00 / 65.00 mm

Clevis pin bore 20; nominal ear gap 56 mm.

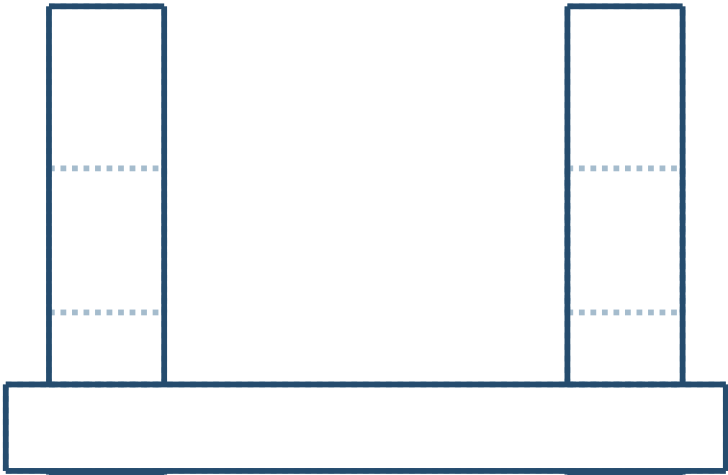
Foot 4 x diameter 9 at (+/-20,+/-35); base mating holes pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

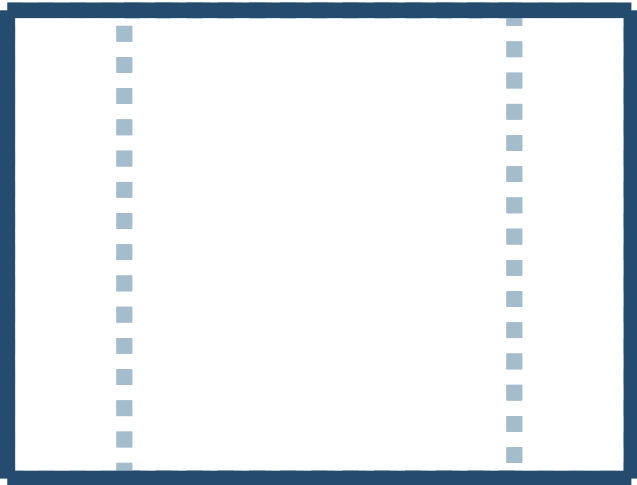
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 32.00 / 24.00 / 290.00 mm

32 x 24 section; pin centres Z 0, 205, 260 in local coordinates.

Nominal pin bore 20; bush fit not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



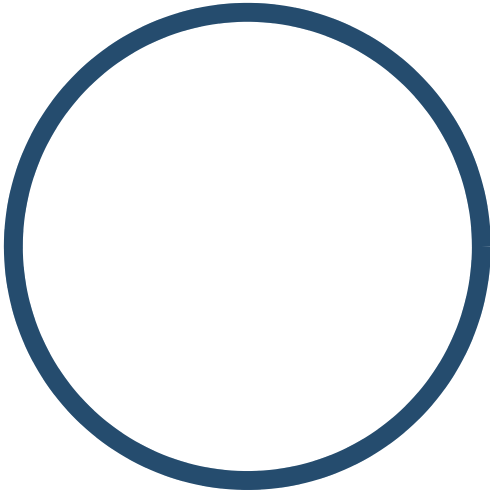
AB-DR-010 / Lower pivot pin 1

AI BRAKING / AB-DR / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind; specify retention after review

Overall X/Y/Z bounds: 19.80 / 105.00 / 19.80 mm

Diameter 19.8 study clearance in nominal 20 bore.

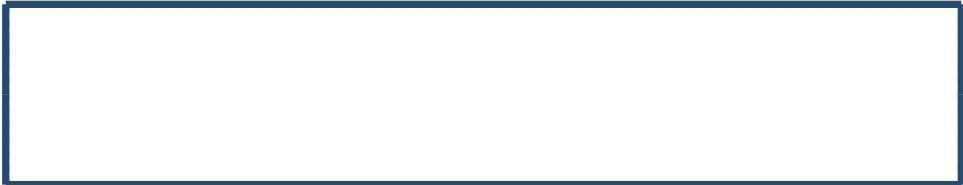
No final fit, hardness or axial retainer design.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-DR-011 / Lining 1

AI BRAKING / AB-DR / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 36.03 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

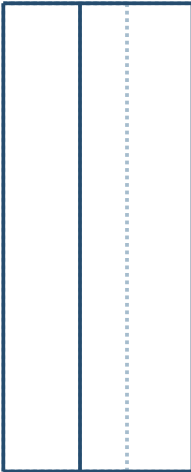
SIDE Y-Z



AB-DR-012 / Shoe carrier 1

AI BRAKING / AB-DR / C0 / 2026-09-07

TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 40.42 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

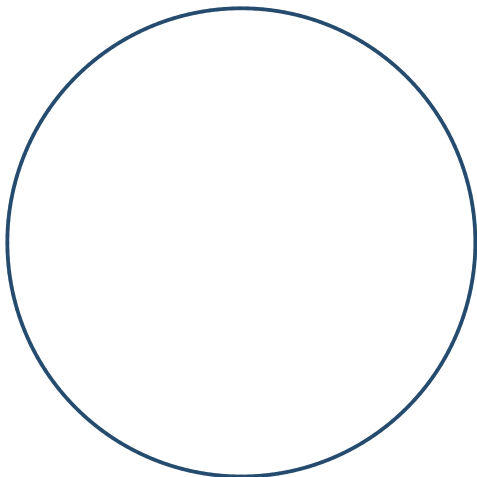
SIDE Y-Z



AB-DR-013 / Thruster reserved space

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy selected thruster; replace with vendor STEP

Overall X/Y/Z bounds: 100.00 / 100.00 / 190.00 mm

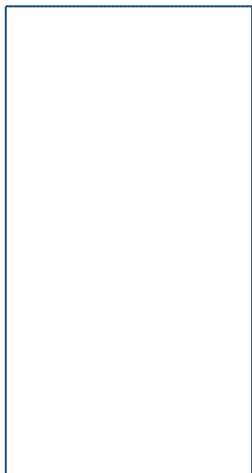
Nonfunctional body envelope; linkage not included.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z



AB-DR-014 / Spring/equalizer reserved space

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Engineer spring/linkage after kinematics

Overall X/Y/Z bounds: 160.00 / 50.00 / 45.00 mm

Reserved volume only; no spring, force path or release motion solved.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

